AI ASSISTED CODING

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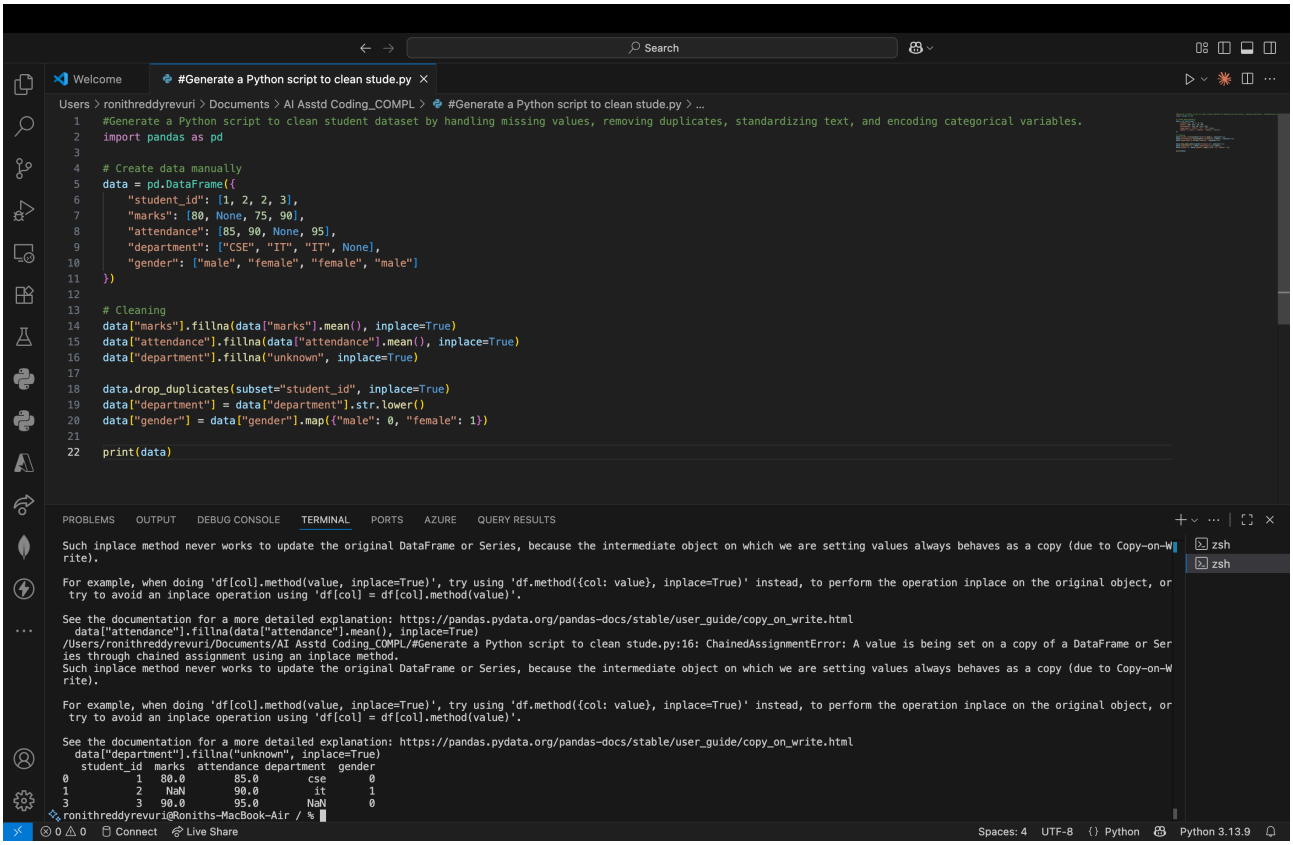
TASK - 1 : 1(STUDENT ACADEMIC DATA CLEANING)

PROMPT :
GENERATE A PYTHON SCRIPT TO CLEAN STUDENT DATASET BY HANDLING MISSING VALUES, REMOVING DUPLICATES, STANDARDIZING TEXT, AND ENCODING CATEGORICAL VARIABLES.

EXPLANATION :

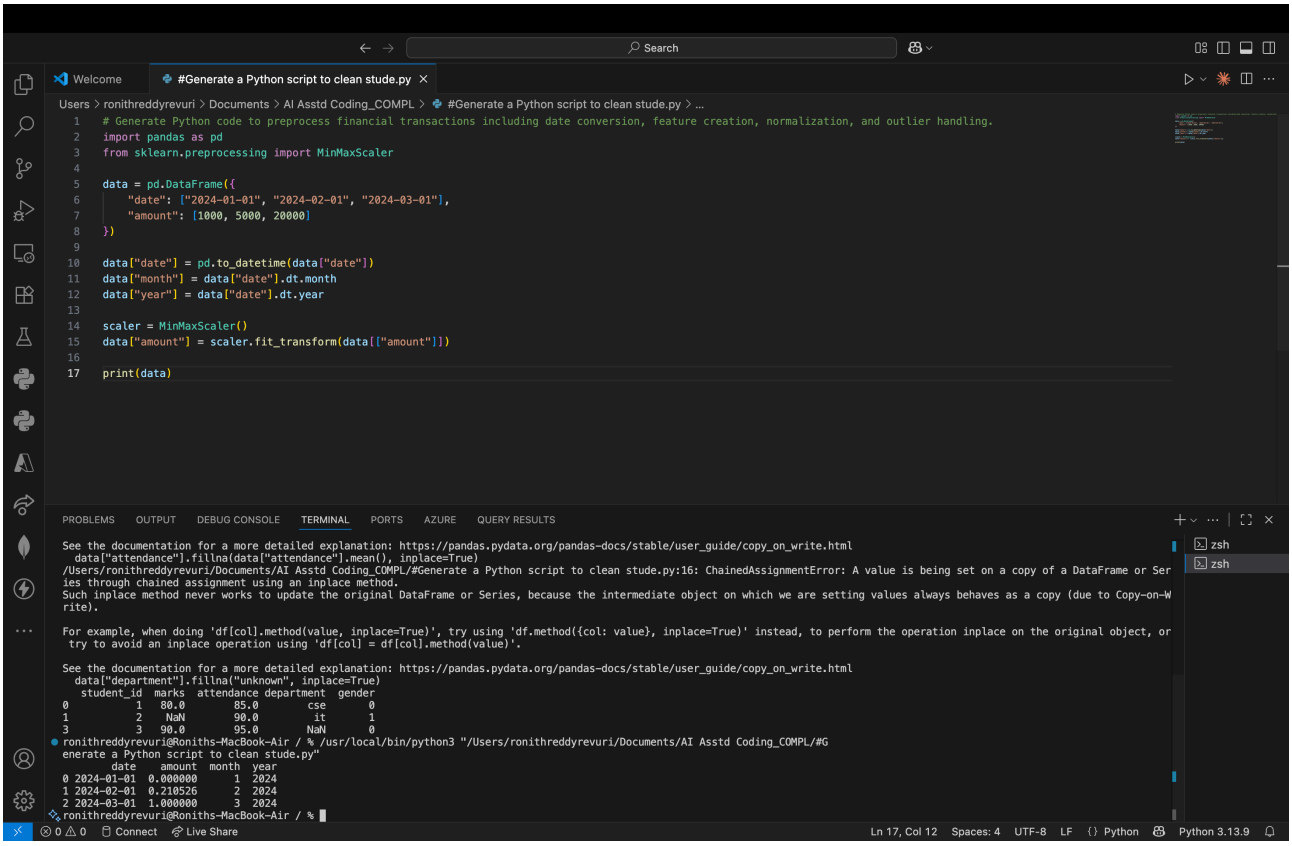
THIS TASK CLEANS STUDENT DATA BY FILLING MISSING VALUES AND REMOVING DUPLICATE RECORDS. IT ALSO STANDARDIZES TEXT LIKE DEPARTMENT NAMES AND CONVERTS CATEGORICAL DATA (LIKE GENDER) INTO NUMBERS. THIS MAKES THE DATASET READY FOR ANALYSIS.



TASK 2 : (FINANCIAL TRANSACTION DATA PREPROCESSING)

PROMPT :
GENERATE PYTHON CODE TO PREPROCESS FINANCIAL TRANSACTIONS INCLUDING DATE CONVERSION, FEATURE CREATION, NORMALIZATION, AND OUTLIER HANDLING.

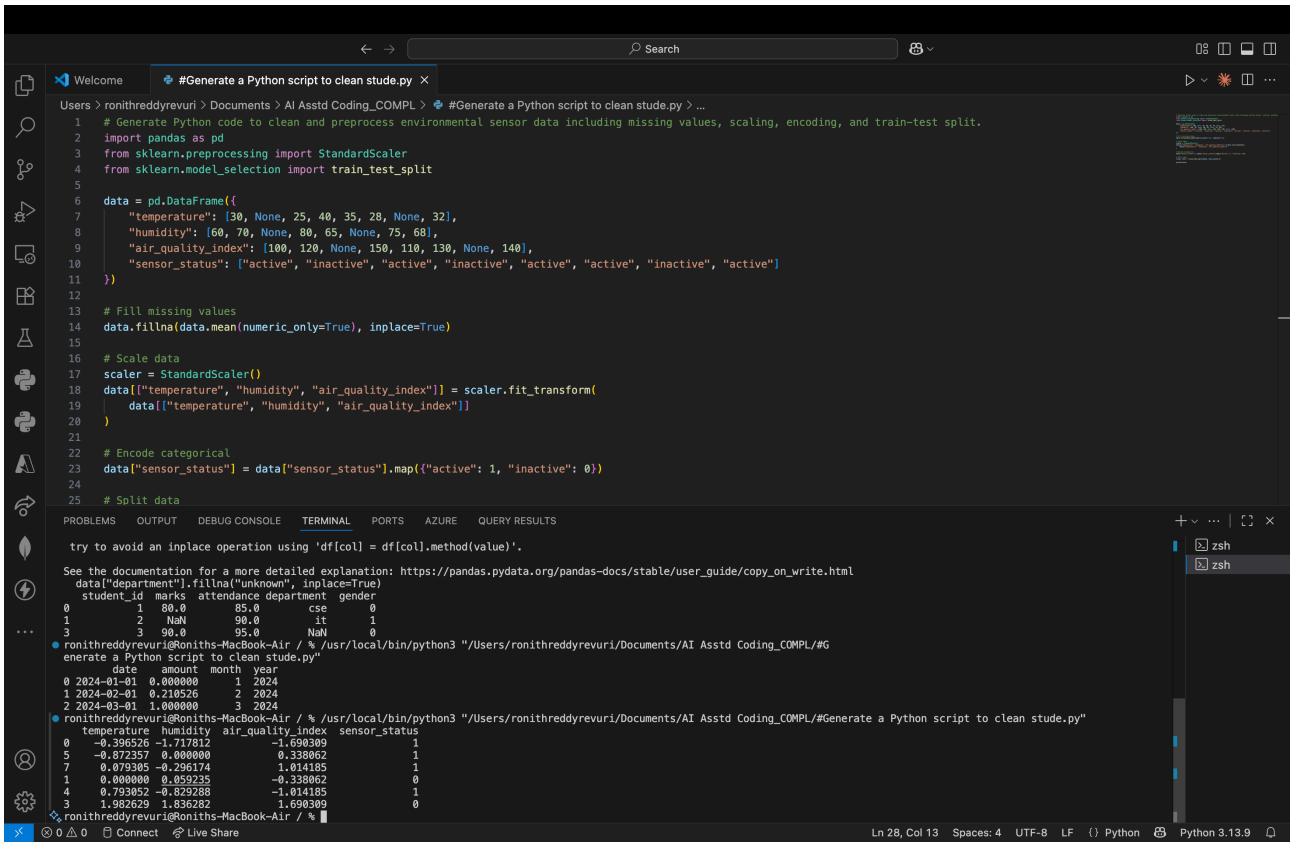
EXPLANATION :
IN THIS TASK, THE TRANSACTION DATA IS IMPROVED BY CONVERTING DATES INTO A PROPER FORMAT AND EXTRACTING USEFUL DETAILS LIKE MONTH AND YEAR. THE AMOUNT VALUES ARE NORMALIZED AND EXTREME VALUES ARE HANDLED. THIS HELPS IN BETTER FINANCIAL ANALYSIS.



TASK 3 – (ENVIRONMENTAL SENSOR DATA PREPARATION)

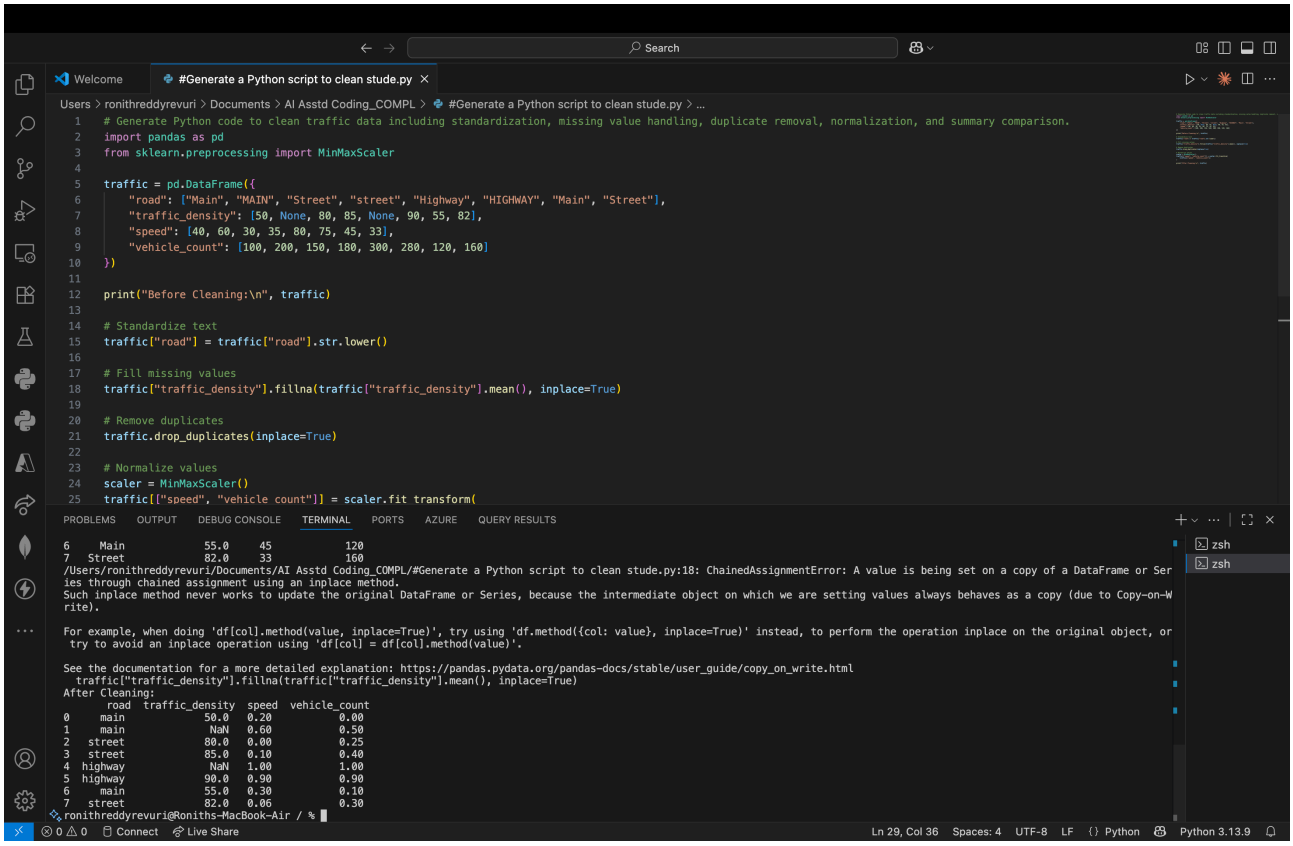
PROMPT : GENERATE PYTHON CODE TO CLEAN AND PREPROCESS ENVIRONMENTAL SENSOR DATA INCLUDING MISSING VALUES, SCALING, ENCODING, AND TRAIN-TEST SPLIT.

EXPLANATION :
THIS TASK PREPARES SENSOR DATA BY FILLING MISSING VALUES AND SCALING NUMERICAL DATA FOR CONSISTENCY. IT ALSO CONVERTS STATUS LABELS INTO NUMBERS AND SPLITS THE DATA FOR TRAINING AND TESTING. THIS MAKES THE DATA READY FOR MACHINE LEARNING MODELS.



TASK 4 : (REAL-TIME APPLICATION: SMART CITY TRAFFIC DATA CLEANING)
PROMPT :
GENERATE PYTHON CODE TO CLEAN TRAFFIC DATA INCLUDING STANDARDIZATION, MISSING VALUE HANDLING, DUPLICATE REMOVAL, NORMALIZATION, AND SUMMARY COMPARISON.

EXPLANATION :
TRAFFIC DATA IS CLEANED BY STANDARDIZING NAMES AND REMOVING DUPLICATE RECORDS. MISSING VALUES ARE FILLED AND IMPORTANT VALUES ARE NORMALIZED. THE DATA IS ALSO COMPARED BEFORE AND AFTER CLEANING TO SHOW IMPROVEMENT.



TASK - 5 : (SOCIAL MEDIA TEXT DATA CLEANING AND FEATURE EXTRACTION)
PROMPT :
GENERATE PYTHON CODE TO CLEAN SOCIAL MEDIA TEXT DATA BY REMOVING DUPLICATES, CLEANING TEXT, TOKENIZATION, STOPWORD REMOVAL, AND FEATURE EXTRACTION.

EXPLANATION :
THIS TASK CLEANS TEXT DATA BY REMOVING UNWANTED ELEMENTS LIKE LINKS AND SPECIAL CHARACTERS. IT CONVERTS TEXT TO LOWERCASE AND REMOVES UNNECESSARY WORDS. FINALLY, IT EXTRACTS USEFUL FEATURES LIKE WORD COUNT FOR ANALYSIS.

Users > ronithreddyrevuri > Documents > AI Asstd Coding_COMPL > #Generate a Python script to clean stude.py > ...

```python
# Generate Python code to clean social media text data by removing duplicates, cleaning text, tokenization, stopword removal, and feature extraction.

import pandas as pd
import re

posts = pd.DataFrame({
    "text": [
        "Hello world!!!",
        "Visit http://example.com",
        None,
        "AI is amazing!!!",
        "HELLO WORLD!!!",
        "Check this link https://test.com",
        "Data Science is fun :)",
        "AI is amazing!!!"
    ]
})

# Remove duplicates and nulls
posts.drop_duplicates(inplace=True)
posts.dropna(inplace=True)

def clean(text):
    text = re.sub(r"http\S+", "", text)  # remove URLs
    text = re.sub(r"[^a-zA-Z ]", "", text)  # remove special chars
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE QUERY RESULTS

```
See the documentation for a more detailed explanation: https://pandas.pydata.org/pandas-docs/stable/user_guide/copy_on_write.html
  traffic["traffic_density"].fillna(traffic["traffic_density"].mean(), inplace=True)
After Cleaning:
       road  traffic_density  speed  vehicle_count
0      main             50.0   0.20           0.00
1      main              NaN   0.60           0.50
2    street             80.0   0.00           0.25
3    street             85.0   0.10           0.40
4   highway              NaN   1.00           1.00
5   highway             90.0   0.90           0.90
6      main             55.0   0.30           0.10
7    street             82.0   0.06           0.30
ronithreddyrevuri@Roniths-MacBook-Air / % /usr/local/bin/python3 "/Users/ronithreddyrevuri/Documents/AI Asstd Coding_COMPL/#G
enerate a Python script to clean stude.py"
                                 text           clean_text  word_count
0                     Hello world!!!          hello world           2
1           Visit http://example.com                visit           1
3                   AI is amazing!!!        ai is amazing           3
4                     HELLO WORLD!!!          hello world           2
5  Check this link https://test.com      check this link           3
6            Data Science is fun :)  data science is fun           4
ronithreddyrevuri@Roniths-MacBook-Air / %
```

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